Data Mining

Data Governance and Collection

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Inconsistent Data Across Systems Can Disrupt Decision-Making

The Situation

Marketing requested the quarterly customer report.
When the data team generated the numbers, the results did not match across systems.
Different departments reported different customer counts and revenue totals.

Which numbers were correct?

What Went Wrong?

- Three databases → three definitions of a customer
- Different data cleaning practices
- No shared ownership of datasets

Result: loss of trust in the data

Organizations Need Governance to Build Trust in Data

Key Insight

Without shared definitions, ownership, and validation, organizations cannot rely on their data.

- Establish a single source of truth
- Align data definitions across teams
- Assign clear data ownership
- Implement automated quality checks

Data Governance Defines How Organizational Data Is Managed

- Framework for managing enterprise data
- Defines policies, standards, and responsibilities
- Ensures data is accurate, secure, and compliant
- Establishes a shared single source of truth

Governance Enables Data Quality and Business Value

Data Governance

- Policies
- Standards
- Ownership
- Stewardship

$\Longrightarrow$

Data Quality

- Accuracy
- Completeness
- Consistency
- Timeliness

$\Longrightarrow$

Business Impact

- Better decisions
- Operational efficiency
- Regulatory compliance
- Customer trust

Five Core Components Enable Effective Data Governance

- Data Quality – ensuring accuracy and reliability
- Data Security – protecting data from unauthorized access
- Data Privacy – compliance with regulations
- Data Stewardship – assigning responsibility for data
- Metadata Management – documenting definitions and lineage

Organizations Often Face Governance Implementation Challenges

- Limited executive support for governance initiatives
- Resistance to change when introducing new policies
- Complex data ecosystems across multiple platforms
- Evolving regulatory requirements requiring ongoing updates

High Data Quality Is Essential for Reliable Analytics

- Organizations rely on data for analytics and decision-making
- Poor data quality leads to incorrect insights
- High-quality data ensures reliable information

Accuracy, Completeness, and Consistency Define Data Quality

$$\text{Accuracy} \Rightarrow \text{Completeness} \Rightarrow \text{Consistency}$$

- Accuracy: data reflects real-world values
- Completeness: required information is present
- Consistency: data is uniform across systems

Accurate Data Reflects Real-World Values

Example

Customer addresses differ between an online store and a physical store CRM.
Incorrect addresses lead to shipping errors.

- Validate data against trusted sources
- Apply data entry validation rules
- Synchronize systems regularly

Complete Data Ensures Information Is Usable

Example

A marketing campaign imports leads into a CRM, but many records lack email addresses.
Follow-up campaigns cannot be executed.

- Require mandatory fields
- Monitor pipelines for missing values
- Use data enrichment tools

Reliable Data Requires Consistent Reporting Practices

Example

Regional offices report financial data using different reporting schedules. This produces inconsistent dashboards.

- Standardize reporting formats
- Conduct regular data audits
- Use automated data integration pipelines

Trusted Data Enables Confident Decision-Making

Key Takeaway

Data governance provides the structure, data quality ensures reliability, and trusted data enables confident decisions.

Reliable Decisions Depend on Reliable Data

- Modern organizations rely on data driven decision making
- Decisions are only as good as the data available
- A data inventory helps identify
  - available datasets
  - information gaps
- Missing information must be obtained through data collection

Data Collection Is a Structured Process

- Data collection gathers information using systematic methods
- The goal is to obtain reliable and meaningful data
- Collected data supports
  - analysis
  - monitoring
  - strategic decisions

Primary and Secondary Sources of Data

Primary data

- collected directly from the source
- examples: surveys, interviews, experiments

Secondary data

- previously collected by other organizations
- examples: reports, archives, public datasets

Primary Methods Capture Data Directly

- Interviews
- Surveys and questionnaires
- Observations
- Focus groups
- Experiments

Existing Sources Provide Valuable Data

- Government datasets
- Research publications and academic databases
- Company reports and industry studies
- Open data portals and public repositories

Modern Technologies Enable Large Scale Data Collection

- Online and mobile surveys
- Sensors and IoT devices
- Application and system logs
- Web scraping tools

Effective Data Collection Follows a Clear Process

1. Define the data requirements
2. Select the appropriate collection method
3. Gather the required information
4. Analyze and interpret the collected data

Data Collection Often Faces Practical Challenges

- Maintaining data quality and consistency
- Detecting and preventing collection errors
- Integrating information from multiple sources
- Managing increasing data volume

Quantitative Data Supports Statistical Analysis

- Quantitative data represents numerical measurements
- Enables statistical analysis and modeling
- Common collection methods include
  - surveys
  - experiments
  - sensor measurements

Experiments and A B Testing Evaluate Alternatives

- Experiments measure the effect of controlled changes
- A B testing compares two alternative versions of a system
- Performance metrics may include
  - conversion rate
  - click through rate
  - engagement

Qualitative Data Captures Context and Meaning

- Qualitative data represents opinions, experiences, and perceptions
- Typical sources include
  - interviews
  - focus groups
  - open ended survey responses
  - observational notes
- Interpretation may involve subjectivity and contextual analysis

High Quality Data Enables Better Decisions

- Effective data collection is essential for data driven activities
- Both quantitative and qualitative data contribute insights
- Choosing appropriate methods improves data quality
- Reliable data supports innovation and informed decision making

Key Takeaway

Reliable insights begin with reliable data,
and reliable data depends on systematic data collection.